

ASSAYBENCH

1,920

historical CRISPR screens

SK-N-DZ neuroblastoma
CRISPRa • RSL3, 300 nM
ferroptosis resistance

KBM-7 leukemia
CRISPRko • etoposide, 12 days
drug-resistance phenotype

HEK-293 kidney cells
CRISPRko • IL-1beta, 8 days
increased NF-kB reporter signal

...

BIOLOGY-IN-THE-LOOP

Which genes, when activated in neuroblastoma,
confer resistance to RSL3?

predict hit genes

ASSAYLOOP

AssayLLM

AssayFormer

observe hit labels

100 genes / round • 10 rounds

ASSAYBENCH-LOOP

Sequential hit-discovery benchmark

chronological out-of-distribution split

1,349 TRAIN
<2021

20 VAL
2021

20 TEST
>2021

5 phenotype families

- fitness / viability
- drug / environment
- host-pathogen
- molecular output
- trafficking / structure

8 evaluation metrics

1,000 gene budget

missing-gene aware

From historical CRISPR screens to a sequential experimental-design benchmark